Metaboenrich step 5 (experimental): a directed absorbing random walk conditioned on absorption

Metaboenrich methods note

Code reference: `metaboenrich/walk.py`, commit `0f6c135` (branch `directed-rw`); option `-mode walk`

Abstract

The default Metaboenrich scoring step, Ohm's-law current flow, solves an electrical problem on the *undirected* metabolite–reaction graph. It ignores reaction directions and lets current pass between two substrates of the same reaction. This note describes the experimental alternative. Walkers are released at the metabolites that went up, in proportion to $|\log_2 \mathrm{FC}|$, and step only along reaction directions until a metabolite that went down absorbs them. Walkers that reach a dead end are lost, and only walks that reach a target are scored, using Doob's h-transform. One sparse LU factorisation gives every quantity. The output has the same form as current flow, so ranking and outputs are shared. The method is an absorbing Markov chain. It is not a random walk with restart: walkers never jump back to their start, and scores are not a stationary distribution.

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1 Where this step sits in the pipeline

With `-mode walk`, steps 1–4 of the main run are unchanged (load Human-GEM, currency rules, DAA mapping, graph). Step 5 calls `walk.absorbing_walk` instead of `flow.current_flow`. It returns the same `FlowResult`, so step 6 (scoring) and step 7 (outputs) are shared. The network is the same locked network (3,958 metabolites, 6,377 reactions, 16,356 edges).

Why change step 5. Current flow (companion note) has three properties that clash with biochemistry:

1. it is direction-blind: current can run backwards through an irreversible reaction;
2. it allows co-substrate hops: for $A + B \to C$, current can go $A \to r \to B$;
3. fold change does not enter the electrical problem.

The walk addresses all three. It also replaces "one unit per source–target pair" with "walkers per source, absorbed by whichever target they reach first".

2 Inputs and notation

- $G = (V, E)$: the bipartite metabolite–reaction graph, $V = M \cup R$. Each edge $e = (\mu, \rho)$ joins a metabolite μ to a reaction ρ and is labelled *substrate* or *product*.
- Reaction directions from the Human-GEM flux bounds: ρ can run forward if its upper bound is > 0 and backward if its lower bound is < 0. Reversible reactions can do both.
- $\mathcal{S} \subset M$: sources (significant, $\log_2 \mathrm{FC} > 0$). $\mathcal{T} \subset M$: targets (significant, $\log_2 \mathrm{FC} < 0$).
- $\varphi_v = |\log_2 \mathrm{FC}_v|$ for every source and target.
- $\kappa \geq 0$ (optional, `-walk-kappa`): partial-absorption strength.

3 The state space

A walker on a metabolite must remember which way it is travelling, or it could leave a reaction through the side it entered. The walk therefore runs on a directed graph of *states* $\mathcal{X}$, built from G (`walk_states`).

States.

- *Start states*: one per metabolite μ, where walkers released at μ begin.
- *Direction states*: $\rho^{\rightarrow}$ if ρ can run forward, $\rho^{\leftarrow}$ if it can run backward. A reversible reaction has two.
- *Arrival states*: one per step "direction d produces metabolite μ", written $(\mu \mid d)$: the walker is on μ and arrived via d.

Arcs. Write $\mathrm{in}(d)$ for the metabolites a direction consumes and $\mathrm{out}(d)$ for those it produces: substrates and products for $\rho^{\rightarrow}$, the reverse for $\rho^{\leftarrow}$.

- start $\mu \to d$ for every direction d with $\mu \in \mathrm{in}(d)$;
- $d \to (\mu \mid d)$ for every $\mu \in \mathrm{out}(d)$;
- $(\mu \mid d) \to d'$ for every d' with $\mu \in \mathrm{in}(d')$, **except** $d' = \overline{d}$, the opposite direction of the same reaction.

The first two rules mean a walker enters a reaction through what it consumes and leaves through what it makes. The third, the *no-immediate-reversal rule*, stops the two-step co-substrate hop $A \to \rho^{\rightarrow} \to C \to \rho^{\leftarrow} \to B$ through a reversible reaction. Each arc runs along one edge of G and carries a sign: $+1$ for metabolite $\to$ reaction, -1 for reaction $\to$ metabolite. On Human-GEM, $|\mathcal{X}| = 21{,}787$ with 179,703 arcs.

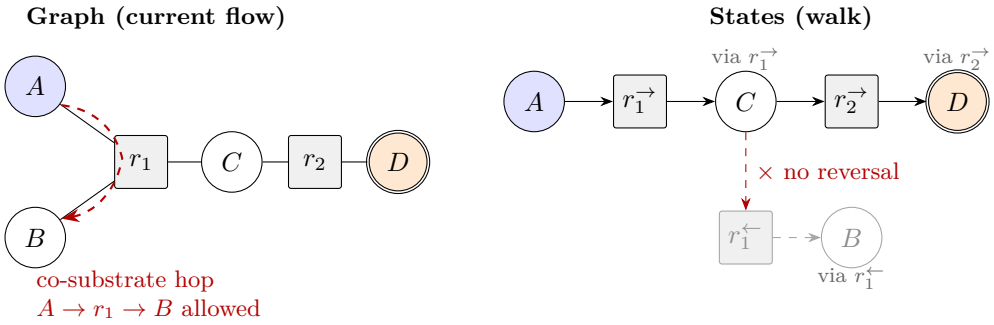


Figure 1: Toy network $r_1 : A + B \rightleftharpoons C$ (reversible), $r_2 : C \to D$; source A, target D. Left: on the undirected graph, current from A can leave r_1 through its co-substrate B. Right: in the walk's state space, a walker from A goes $r_1^{\rightarrow} \to C \to r_2^{\rightarrow} \to D$. Returning through $r_1^{\leftarrow}$ to B is forbidden because $r_1^{\leftarrow}$ is the opposite of the step just taken.

4 Absorption, leaks and the transition matrix

Absorption (where the DAA enters, hook 2). Only *arrival* states of targets absorb: a walker must reach a target through a reaction. For an arrival state $\alpha = (t \mid d)$ with $t \in \mathcal{T}$,

$$a_\alpha = \begin{cases} 1 & \text{full absorption (default),} \\ 1 - e^{-\kappa \varphi_t} & \text{partial absorption, } \kappa \text{ given.} \end{cases} \tag{1}$$

All other states have $a_u = 0$. With partial absorption a deep drop catches almost every walker, while a mild one lets some walk on to targets further along.

Transitions. From state u with out-degree $\deg^+(u)$ in $\mathcal{X}$, a walker that is not absorbed moves along one of its arcs uniformly at random:

$$Q_{uv} = \frac{1 - a_u}{\deg^+(u)} \quad \text{for each arc } u \to v. \tag{2}$$

Leaks. A state with no way out (a dead end) loses its walkers. More generally, let $\mathcal{X}_{\text{live}}$ be the states from which some absorbing state is reachable (reverse breadth-first search, `_can_reach`). Only arcs between live states are kept in Q, but $\deg^+(u)$ still counts *all* arcs, so probability mass

that steps into a dead region is lost. Pruning dead states does not change any result below, because a walk that enters them can never be absorbed.

5 Absorption probabilities and expected visits

Restrict everything to $\mathcal{X}_{\text{live}}$ and let I be the identity there.

Probability of reaching a target. Let $h(u)$ be the probability that a walk from state u is eventually absorbed. First-step analysis gives $h = a + Qh$, so

$$(I - Q)\, h = a. \tag{3}$$

For a source s, $h(s)$ is its *reach probability*. Sources with $h(s) = 0$ are excluded ("no directed path to a target").

Expected visits. The fundamental matrix $N = (I - Q)^{-1}$ has N_{su} = expected number of visits to u by a walk started at s. Its row for s is

$$n_s = (I - Q)^{-\top} e_s. \tag{4}$$

Both (3) and (4) use the same LU factorisation of $I - Q$: (4) is a transposed solve.

6 Conditioning on success: Doob's h-transform

Most released walkers never reach a target: in the example run only about 20% do, the rest end in directed dead ends. Scoring all walkers would reward reactions that lead into dead ends. The walk therefore scores only the walks that are absorbed.

Proposition. *Conditioned on absorption, the walk is again a Markov chain, with transitions* $\widehat{Q}_{uv} = Q_{uv}\, h(v)/h(u)$ *(Doob's h-transform). For a walk from source* s*, the expected number of conditioned crossings of arc* $u \to v$ *is*

$$c_s(u \to v) = \frac{n_s(u)\, Q_{uv}\, h(v)}{h(s)}, \tag{5}$$

and the probability that a successful walk from s *ends at target* t *is*

$$p_s(t) = \frac{1}{h(s)} \sum_{\alpha=(t|d)} n_s(\alpha)\, a_\alpha. \tag{6}$$

Sketch. The expected number of visits to u by successful walks is $n_s(u)h(u)/h(s)$ (visit u, then succeed from u; divide by the success probability). Multiplying by $\widehat{Q}_{uv}$ gives (5). Equation (6) counts absorptions at t's arrival states among successful walks. □

Under conditioning, each source sends exactly one unit of flow to the targets: $\sum_t p_s(t) = 1$. This puts the walk on the same footing as one unit of current in current flow.

7 Folding traffic back onto the graph

Net edge flow. Sum the signed arc traffic onto the edges of G:

$$F_s(e) = \sum_{\text{arcs } u\to v \text{ along } e} \operatorname{sign}(u \to v)\, c_s(u \to v),$$

positive when, on balance, walkers move from the metabolite into the reaction. Traffic that runs both ways along an edge cancels.

Throughput. As in current flow,

$$T_s(v) = \tfrac{1}{2}\sum_{e\ni v} |F_s(e)| \;-\; \tfrac{1}{2}\, p_s(v)\, \mathbf{1}[v \in \mathcal{T}], \qquad T_s(s) = 0, \qquad T_s(v) \leftarrow \max\{T_s(v), 0\}. \tag{7}$$

A walker absorbed at a target enters but does not leave, so half of its absorbed share is removed: a target scores only the flow it relays onwards. A source's own walkers are not counted at that source.

Direction of use. For a reaction ρ,

$$\text{net_forward}_s(\rho) = \sum_{\text{substrate edges } e \text{ of } \rho} F_s(e),$$

positive when ρ is used forward on balance and negative when used backward.

8 Combining sources into scores

Source weights (where the DAA enters, hook 1). Walkers are released at source s in proportion to φ_s. The share of all *successful* walkers that came from s is

$$w_s = \frac{\varphi_s\, h(s)}{\sum_{s'\in\mathcal{S}} \varphi_{s'}\, h(s')}, \qquad \textstyle\sum_s w_s = 1. \tag{8}$$

Scores. With the reached sources $\mathcal{S}^* = \{s : h(s) > 0\}$:

$$\text{conductivity} \quad c(v) = \textstyle\sum_{s\in\mathcal{S}^*} w_s\, T_s(v) \quad \text{(also reported as accumulated current)}, \tag{9}$$

$$\text{bottleneck} \quad \beta(v) = \max_{s\in\mathcal{S}^*} T_s(v), \tag{10}$$

$$\text{bottleneck fraction} \quad f_\tau(v) = \textstyle\sum_s w_s\, \mathbf{1}[T_s(v) \ge \tau], \quad \tau = 0.5, \tag{11}$$

$$\text{edge traffic} \quad \bar{F}_e = \textstyle\sum_s w_s |F_s(e)|, \qquad F_e^{\max} = \max_s |F_s(e)|, \tag{12}$$

$$\text{net forward} \quad \mathrm{nf}(\rho) = \textstyle\sum_s w_s\, \text{net_forward}_s(\rho). \tag{13}$$

Leave-one-out scores are not defined for the walk; those columns are empty and `-scoring loo` is refused.

Reported per metabolite and per run. For a source, the `eff_conductance` field holds its reach probability $h(s)$; for a target, its share of successful walkers $\sum_s w_s p_s(t)$. A pair (s, t) is "reached" if $p_s(t) > 0$; others are excluded ("no walk reaches this target"). The run summary adds the success fraction $\sum_s \varphi_s h(s) / \sum_s \varphi_s$, the number of unreached sources, and the state, live-state and arc counts.

9 One-sided data

If only sources are significant, the measured but unchanged metabolites (the ground) absorb every walker ($a = 1$ at their arrival states). If only targets are significant, walkers start evenly from the ground ($\varphi = 1$ each). Ground nodes are not scored as endpoints.

10 From node scores to ranked genes and pathways

Identical to current flow (`scoring.py`): a reaction's final score is the mean percentile rank of c and β; a gene takes its best reaction; pathways are tested by hypergeometric over-representation among the top 100 reactions with Benjamini–Hochberg correction. `reactions.tsv` gains the `net_forward` column. The Cytoscape subnetwork keeps edges with $F_e^{\max} \geq 0.1$.

11 Computation

1. Build $\mathcal{X}$ and its arcs once (21,787 states on Human-GEM).
2. Reverse BFS from absorbing states; keep live states (16,951 in the example run).
3. One sparse LU of $I - Q$ (`scipy.sparse.linalg.splu`), about 0.3 s.
4. Solve (3) once; solve (4) for blocks of 128 sources.
5. Sparse products fold traffic onto edges and nodes.

That is $1 + |\mathcal{S}|$ solves, against $|\mathcal{S}| + |\mathcal{T}|$ for current flow.

12 Properties and limits

- **Respects direction.** Irreversible reactions are used only forward; reversible ones either way, recorded in `net_forward`.
- **No co-substrate hops**, including the two-step hop through a reversible reaction.
- **Uses fold change** as release weights, and optionally as absorption strength.
- **Targets compete.** A walker stops at the first target that absorbs it, so with full absorption a target behind another target receives little. Partial absorption (κ) relaxes this; on the benchmark it changed little.
- **Throughput can exceed 1.** Directed cycles of irreversible reactions can circulate walkers before they leave.
- **Sensitive to direction annotations.** A reaction marked irreversible in the wrong direction blocks every route through it. If the measured metabolites point away from the block, as with shunt products, walkers cannot reach it.
- **Only successful walks count.** When few walkers reach a target (low success fraction), scores rest on a small share of the released walkers.

Remark (Relation to current flow and to RWR)**.** *On an undirected graph, unit current from s to t equals the expected net edge crossings of a simple random walk from s absorbed at t (Newman, 2005). The walk here changes three things: it is directed (on the state space), all targets absorb together, and sources are weighted by fold change and conditioned on success. It is not a random*

walk with restart (RWR): an RWR walker returns to its seed with a fixed probability at every step and the score is its stationary visit frequency, whereas here walks end at a target or a dead end and the score is the traffic of successful walks.

13 Validation

`tests/test_walk.py` checks the rules on hand-built networks: a chain carries 1; a backward-only reaction blocks the forward route and is used backward (net forward = −1); no co-substrate hop, even in two steps; a dead-end branch leaks half the walkers and scores 0; partial absorption splits as $1 - e^{-\kappa\varphi}$ predicts; $\varphi = 3$ vs. 1 splits conductivity 0.75/0.25; one-sided ground absorption. A Monte Carlo simulation (40,000 walkers per source) on a network with branches, a reversible reaction, a directed cycle, a dead end and partial absorption agrees with the exact solution (maximum difference 0.005).

14 Benchmark

Inborn errors of metabolism: rank of the causal gene among 2,302 network genes, raw scoring, full absorption; lower is better. Source: `docs/decisions.md`, D17.

Level	Measure	Current flow	Absorbing walk
Group (20 primary disorders)	median rank	88.5	48
	in top 100	11	13
	in top 10	3	5
Patient (159 primary patients)	median rank	134	67
	in top 100	47%	55%
	in top 10	5%	28%

The walk ranks the causal gene better for 71% of patients and in 14 of 20 disorders. Largest group-level gains: MSUD 33 → 1, PKU 186 → 16, propionic acidaemia 1,187 → 176, citrullinaemia 82 → 20. Losses show where direction hurts. In AADC deficiency (88 → 1,317) the main source, 3-O-methyldopa, is made irreversibly *from* the blocked substrate L-DOPA, so walkers cannot run back to the block. In homocystinuria (181 → 765) the only target is cortisol, which carries no information about the block. In both, fewer than 4% of walkers reach a target. Partial absorption barely matters (group medians 45.5–47 for $\kappa = 0.5, 1, 2$, against 48).

Status. Not the default. It changes protocol §2.3, the evidence is one benchmark, and the blind predictions (D14) were made with current flow. To be decided before the next blind round. Not yet tried: biased steps (hook 3), unchanged metabolites as leaks (hook 4), and the reverse "activation" orientation.

15 Implementation map

Code (`metaboenrich/walk.py`)	What it computes
`walk_states(graph)`	state space $\mathcal{X}$, arcs, arc edges and signs
`_can_reach(...)`	live states $\mathcal{X}_{\text{live}}$
`absorbing_walk(...)`: `absorb`, `q`	a (1), Q (2)
`splu`, `lu.solve(absorb)`	h (3), weights (8)
`lu.solve(E, trans="T")`	visits n_s (4)
`C`, `absorbed`	traffic (5), target shares (6)
`M @ C`, `absB_T`, `S_fwd`	F_s, throughput (7), net forward
`FlowResult`, `walk_info`	same container as current flow, plus run numbers

Command line and benchmark:

```
python -m metaboenrich -daa ...  -mode walk [-walk-kappa K]
python -m metaboenrich.benchmark -method walk [-walk-kappa K]
```

References

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